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IoT shaping agriculture

Things to consider before developing the smart farming solution

Speaker

Vijay (Vijayakumar Manjunatha)

Expert, UN/CEFACT

India

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Modern Agriculture

- ▶ Agriculture was the key development in the rise of human civilization, whereby farming created food surpluses that enabled people to live in cities.
- ▶ Major types: Cultivation of Land & Breeding of Animals / Animal Husbandry
- ▶ History:
 - ▶ 105,000 years ago: Wild grains gathering
 - ▶ 11,500 years ago: Planting / Domesticated grains
 - ▶ 10,000 years ago: Domesticated animals
- ▶ Traditional methods have been improved often with application of new technologies and information.
- ▶ Major Success Parameters: Improved Soil Fertility, Effective Irrigation, Advanced Harvesting, Enhanced Livestock Yield, Crop protection
- ▶ Results in Efficient Produce to Market to Consumption
- ▶ IoT strengthens modern day agricultural goals

Why IoT for agriculture

- ▶ An expected growth of population to 8 billion by 2025 (FAO - Food and Agriculture Organization Prediction).
 - ▶ IoT techniques **enhance the produce** quantity and quality with **reduced cost** and **precision**.
- ▶ Effects of climate change will have greater impact on agriculture (IPCC - UN Panel on climate change prediction).
 - ▶ IoT based methods provide improved irrigation techniques and help decide the **right quantity** and **right time** for water.
- ▶ Various studies have indicated improvisation of agriculture by modern techniques over past century.
 - ▶ IoT gives a major leap to this with direct benefit of **technology** and **information**.
- ▶ Automation has been the main driver in meeting the rising demands in agriculture.
 - ▶ IoT drives automation to next level with **sophisticated decision making** by use of **connectivity and data**.

A few use cases implemented for IoT

▶ Monitoring

- ▶ Basic cameras for monitoring the field / livestock.
- ▶ Intelligent cameras - motion detection with buzz / alarm, smart fishery.
- ▶ Aerial Monitoring systems using drone

▶ Irrigation

- ▶ Remotely operated irrigation systems
- ▶ Smart irrigation controllers (Sensors / Forecasts / Plant-care information)
- ▶ Soil moisture / leaf wetness / Temperature based automation

▶ Automation

- ▶ Aerial planting techniques to sow seeds
- ▶ Pest detection using Infrared sensors
- ▶ Harvesting techniques for crop readiness detection and automation

A few emerging Agriculture use cases

- ▶ Crop protection using Early prediction systems with data crunching
- ▶ Harvesting predictions using customized crop monitoring
- ▶ IR camera and heat sensor based night-predator protection
- ▶ Cattle tracking using GPS based sensors
- ▶ An “Uber” like app for Agri vehicles and equipment
- ▶ Pregnant Cow Monitoring using tail sensors
- ▶ ...and many more

Things to consider on IoT

▶ Major IoT Objectives

▶ Operate

- ▶ Perform specific operations remotely
- ▶ Eg: Irrigation systems, fence activation, Alarm systems, etc

▶ Monitor

- ▶ Data collection and make decisions
- ▶ Eg: Alert mechanisms, early predictions, Harvest decisions, etc

▶ Monitor and Operate

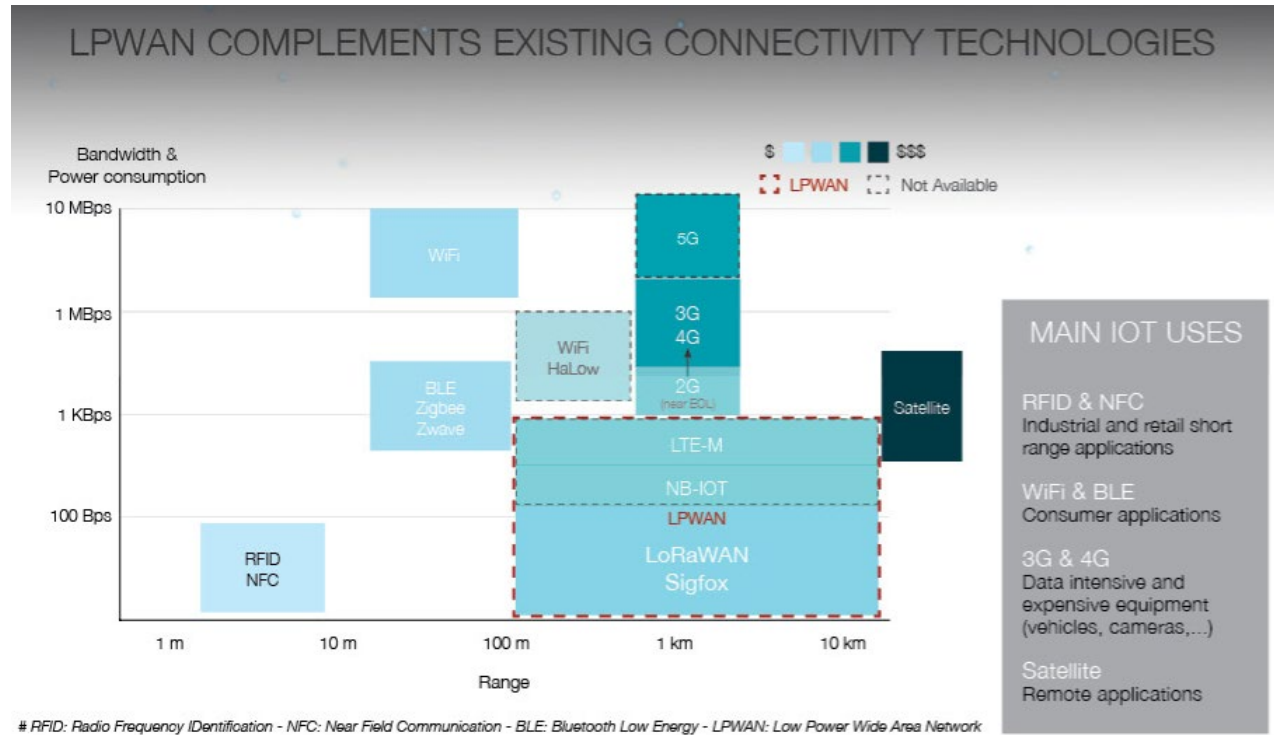
- ▶ Take automated decisions
- ▶ Eg: Soil moisture based irrigation triggers, Camera feed to alarm systems, etc

▶ Coupling to new age systems for robust enhancements

- ▶ Centralized data analysis system
- ▶ Block chain based multi party decision systems
- ▶ Connectivity with business applications

Field Connectivity is the Key

- Connectivity is the key aspect for IoT in order to achieve remote capabilities.



Ref: <https://www.iotforall.com/iot-connectivity-comparison-lora-sigfox-rpma-lpwan-technologies/>

Data is the decision

- ▶ It is important to aggregate data using centralized / distributed platforms.
- ▶ Next leap in smart farming involves data collection and analysis.
- ▶ Community platforms enable right decisions from the historic as well as neighboring patterns.
- ▶ **Social Farming:** Connect and provide Data feed to public or shared systems, Or use of blockchain based decentralized platforms.
- ▶ Implementation of Analytics for early predictions
- ▶ Artificial intelligence for visual data processing coupled with textual data feeds from sensors

Smart Tagging with IoT

- ▶ Production
 - ▶ Farmer / Produce Source
 - ▶ Geography / Region
 - ▶ Factory of Processing
 - ▶ Supply chain participants
 - ▶ {Volume of Supply / Lack of produce}
- ▶ Quality
 - ▶ Quality certifications at various stages
 - ▶ Date of Produce and Life (Age and expiry)
 - ▶ External factors of Agri environment (Rain, Humidity, Dryness, etc)
 - ▶ Fertilizer inputs: Organic, Chemical, etc
- ▶ Consumption
 - ▶ Consumer / Segment
 - ▶ Geography / Region of consumption
 - ▶ {Over-produce / wastage / lack of demand}



BLOCKCHAIN



**ARTIFICIAL
INTELLIGENCE**

International Trade: Produce to Consumption

- ▶ Farmer / Production
 - ▶ Factory / Processing
 - ▶ Packaging
 - ▶ Transport
 - ▶ **Cross border**
 - ▶ Cargo (Air / Ship / Land)
 - ▶ Customs
 - ▶ Distribution & Retail
 - ▶ Customer / Consumption



Need of the hour

- ▶ Sustained Innovation
 - ▶ Practical problem solving
 - ▶ Data distribution & availability
 - ▶ Efficient technologies in hardware and software
 - ▶ Certification mechanisms
 - ▶ Durability of devices due to field implementation
- ▶ Standardization
 - ▶ Protocols and communication
 - ▶ Cross platform and device Interoperability
 - ▶ Extendibility (Current business apps, device to device, etc)
- ▶ Cross Border Acceptance
 - ▶ Mutual recognitions
 - ▶ Global policies / framework
- ▶ Other Areas
 - ▶ Legal framework (Ariel monitoring issues, connectivity networks, etc)

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Thank You!

Vijay (Vijayakumar Manjunatha)

Expert, UN/CEFACT, India

Email: vijay@emudhra.com

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